



Astra Carbon Biochar Methodology Assessment

Isometric – Isometric – Biochar Production and Storage Protocol v1.2

Puro.earth – Biochar 2025 V1

Verra – VCS VM0044 V1.2

22 October 2025

Incentivizing investment in real climate action.

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Introduction

Scope of this report

Sylvera was commissioned by Astra Carbon to conduct an integrity assessment comparison of the following Biochar methodologies:

- [Isometric - Biochar Production and Storage Protocol v1.2](#)¹
- [Puro.earth - Biochar 2025 V1](#)
- [Verra - VCS VM0044 V1.2](#)

This report sets out a **comparative integrity assessment of three leading Biochar methodologies**: Isometric's Biochar Production and Storage Protocol v1.2, Puro.earth's Biochar 2025 V1, and Verra's VCS VM0044. The scope of the work is to evaluate how these methodologies define, quantify, and verify carbon removals, and to highlight where their requirements converge or diverge in ways that may materially affect credit volumes and integrity outcomes of Astra Carbon's project.

The analysis is structured around four key integrity pillars, carbon accounting, additionality, permanence, and safeguards, together with modules on verification requirements and market alignment. By providing a systematic comparison across these dimensions, the assessment offers Astra Carbon a high-level view of the methodological landscape and the relative strengths and risks of each approach.

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⁽¹⁾ Isometric - Biochar Production and Storage Protocol v1.2 is under public consultation until 10/11/2025.

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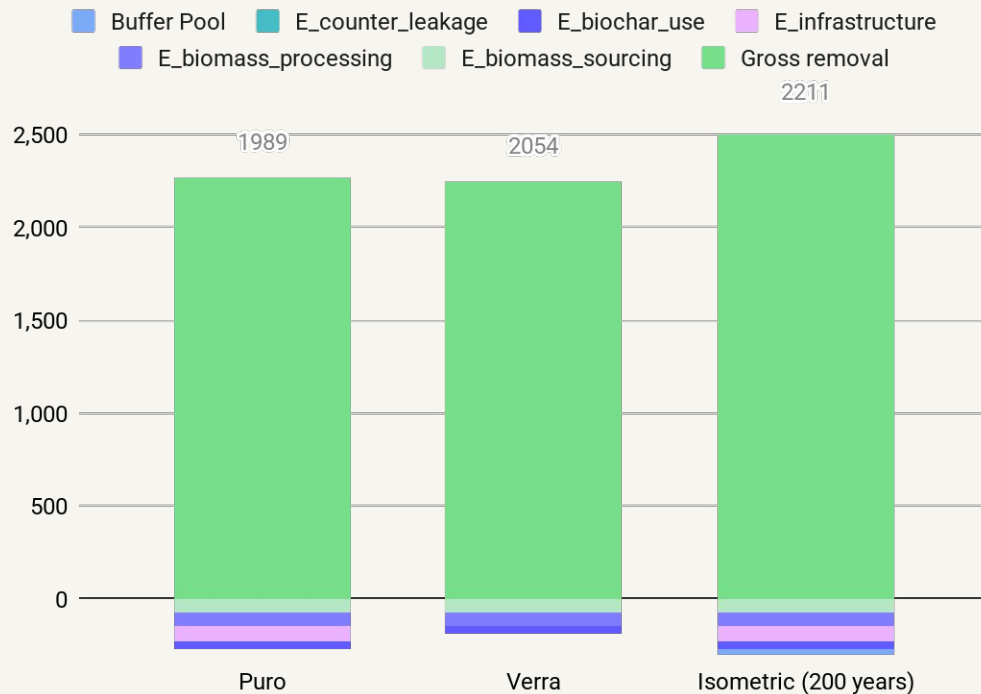
Key takeaways – Module 1 Credit Volume Estimation

Quantification differences drive up to 10% variation in net credit volumes

Credit outcomes are broadly similar across the three methodologies, but underlying assumptions on permanence factor, buffer pool deductions, and treatment of embodied emissions result in meaningful differences as highlighted on the next slide.

Isometric yields the highest net credit volume, driven by its higher permanence factor, despite applying a conservative buffer contribution. In contrast, Verra and Puro apply more conservative assumptions to durable removal potential, resulting in lower overall credit issuance.

Over a 20-year project horizon, cumulative issuance averages 41,700 credits, but the spread between methodologies widens to 4,436 credits by year 20, a 10.6% variation relative to the average.



Key takeaways – Module 2 Methodology Assessment

Isometric and Puro show the lowest risks to integrity as expected due to the CDR-focused and scientifically rigorous approach while Verra displays significant risks across the pillars.

Risk factor	Isometric	Puro	Verra	Commentary
Carbon accounting				Isometric and Puro demonstrate the strongest carbon accounting principles and requirements, mandating transparent and verifiable LCA documentation for emissions and using more stringent scientifically-backed approaches to decay modelling. Verra shows high risks to carbon accounting from non-conservative defaults and assumptions.
Additionality				Approaches to assure additionality are mixed across the three with Isometric showing more robustness in financial additionality testing while Verra and Puro demonstrate more stringent regulatory surplus requirements. While biochar might be deemed uncommon at the time, none of the methodologies are adequately safeguarded against changing practices and regulation.
Permanence				Isometric and Puro have the strongest and most thorough safeguards against carbon stock loss with highly specific end-use considerations. Risks arise across the three when considering post-application land-use change or disturbance, but the overall impact of this risk is reduced by the inherent stability of highly aromatic biochar in durable applications.
Safeguards and co-benefits				All three operate a "do no net harm" policy and score well for community safeguarding. Verra shows the strongest capacity for co-benefits in setting a minimum SDG contribution threshold, while Puro demonstrates the strongest considerations for biochar-specific environmental safeguarding, which is an area where Isometric shows some risk.

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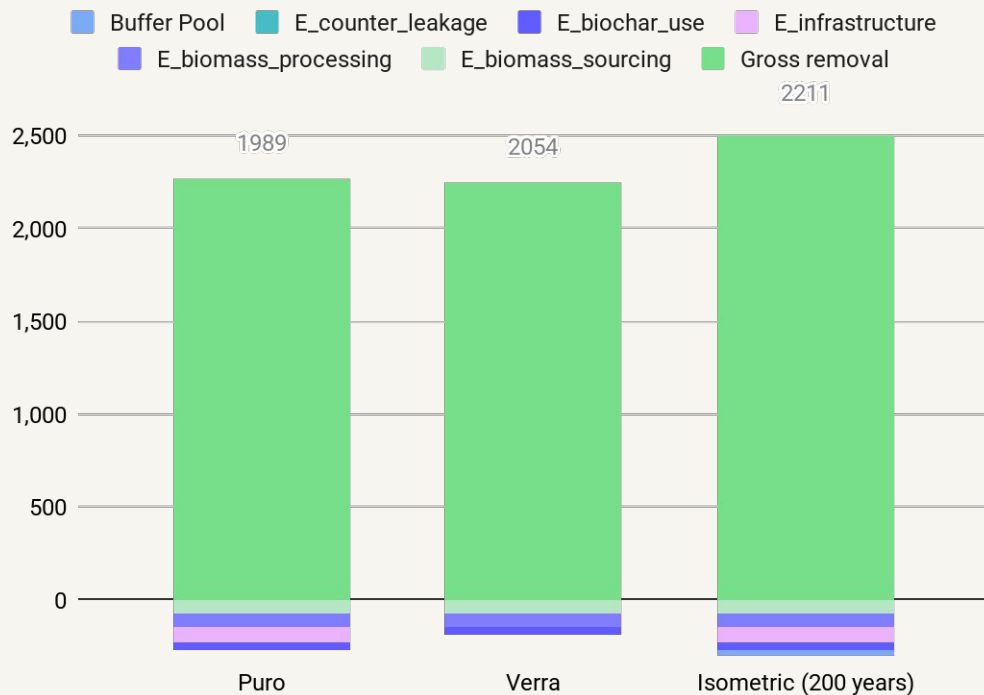
Credit Volume Estimation

Methodologies and Assumptions

- **This analysis is based on a high-level screening life cycle assessment (LCA)**, drawing on literature benchmarks, case studies, and Ecoinvent emission factors. It is intended solely as a guidance tool to understand the likelihood and magnitude of differences between quantification methodologies. It is not a substitute for a full LCA and cannot be used for credit issuance purposes.
- **The greenhouse gas (GHG) sources and sinks considered in this analysis reflect the full life cycle of a biochar project.** These are categorized as follows: (1) gross CO₂ removal resulting from the durable storage of carbon in biochar; (2) emissions from biomass sourcing; (3) emissions from biomass processing; (4) embodied emissions associated with plant construction and equipment; (5) emissions from the transportation and application of biochar; (6) counterfactual emissions and potential leakage related to feedstock use; and (7) buffer pool contributions.
- The project is modeled using a 20-year amortization period, and a hydrogen-to-organic carbon ratio (H/Corg) of 0.200, based on data provided by the developer. In addition, a scenario analysis is included that applies more conservative assumptions.
- Isometric provides both 200-year and 1,000-year permanence modeling options. For consistency across methodologies, the 200-year model is used in this analysis, aligning with Puro.earth's 200-year standard and Verra's 100-year approach. The 1,000-year model offered by Isometric requires Random Reflectance input data, which is not modelled here.
- Feedstock counterfactual and leakage emissions are assumed to be zero, under the premise that the project uses waste-based biomass feedstocks. However, the validity of this assumption is dependent on compliance with feedstock eligibility criteria.
- All emissions are allocated entirely to the biochar carbon credits, with no allocation to co-products or other revenue-generating outputs.

Credit Volume Estimation

Annual Credit Issuance Comparison



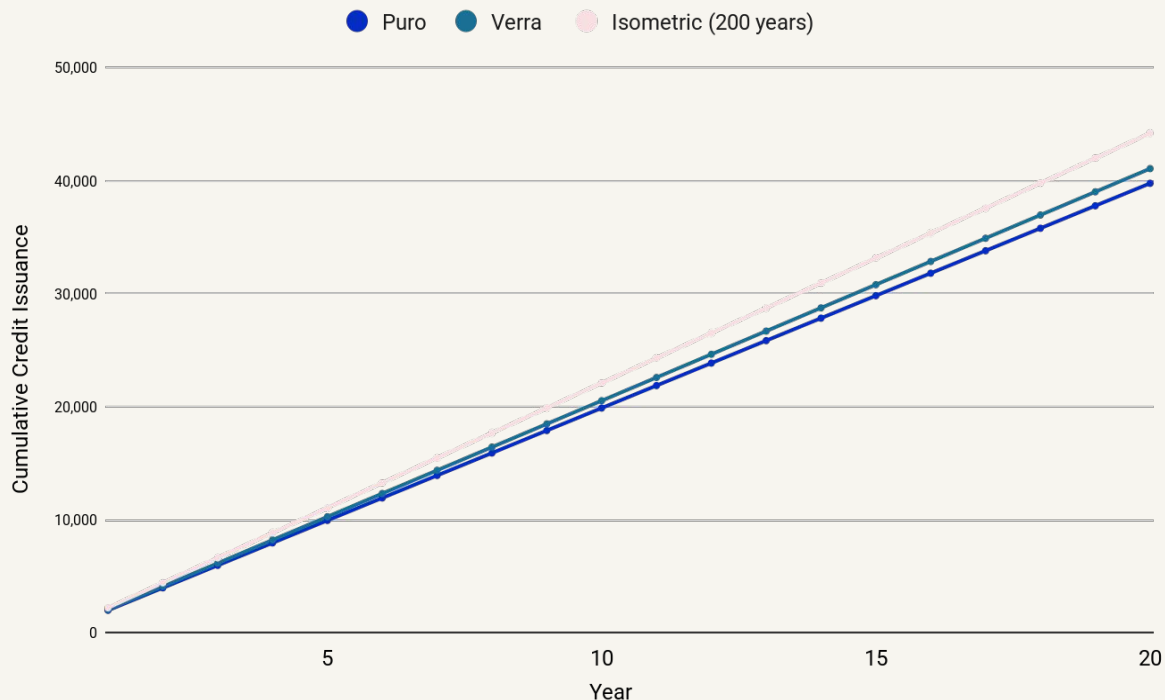
	Puro	Verra	Isometric (200 years)
Gross CO2 removal	2,264	2,242	2,530
E_biomass_sourcing	-69	-69	-69
E_biomass_processing	-75	-75	-75
E_infrastructure	-87	0	-87
E_biochar_use	-44	-44	-44
E_counter_leakage	0	0	0
Buffer Pool	0	0	-45
Net Removal (ton CO2eq)	1,989	2,054	2,211

Key Takeaway:

- **Puro and Verra methodologies produce similar results.** Verra's exclusion of infrastructure emissions slightly inflates credited removals – a potential integrity concern.
- **Isometric yields more credits due to higher gross removal values,** even after applying a 2% buffer deduction.
- Across all three standards, **gross CO₂ removal drives over 85% of total GHG sources and sinks.**

Credit Volume Estimation

Cumulative Credit Issuance Comparison



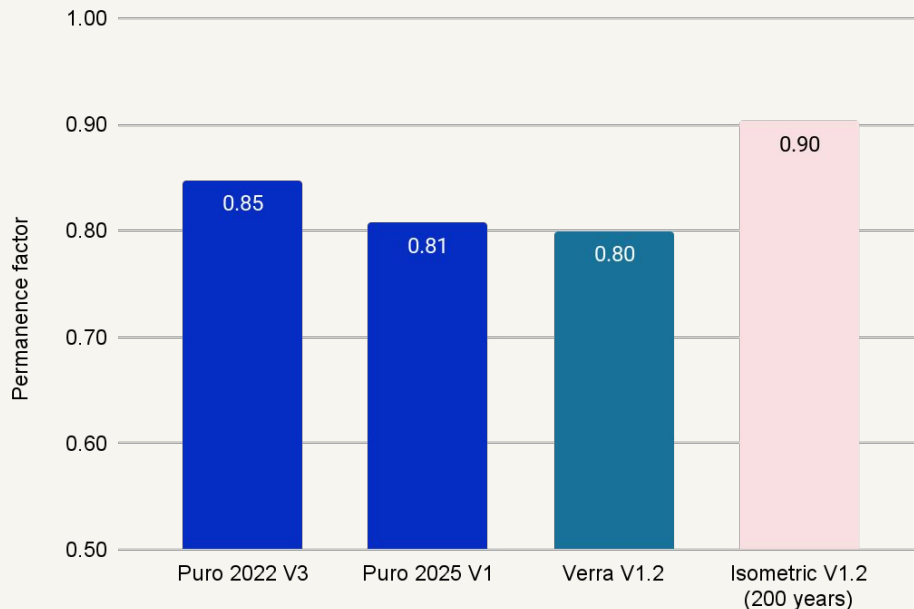
Assuming a consistent biomass input over the 20-year project duration, and an even amortization of embodied emissions across the same period, the cumulative issuance of carbon credits is modeled to increase linearly under all three quantification methodologies.

On average, the project is expected to generate on average **41,688 credits over the 20-year lifespan**.

The difference between the **highest (Isometric)** and **lowest (Puro)** modeled credit outputs amounts to 4,436 credits by year 20, representing a **10.64% variation** relative to the average cumulative issuance.

Credit Volume Estimation

Hotspot Deep-dive: Gross CO₂ Removal and Permanence Factor



Gross CO₂ removal is the dominant factor in credit issuance across three methodologies. It is calculated based on the carbon content of biochar (measured via organic carbon testing), multiplied by the mass produced, the permanence factor, and the molecular weight ratio of CO₂ to carbon (44/12).

The permanence factor adjusts for the fact that biochar degrades over time, with some carbon eventually returning to the atmosphere. In this analysis, differences in permanence factors account for the variation in gross removal values.

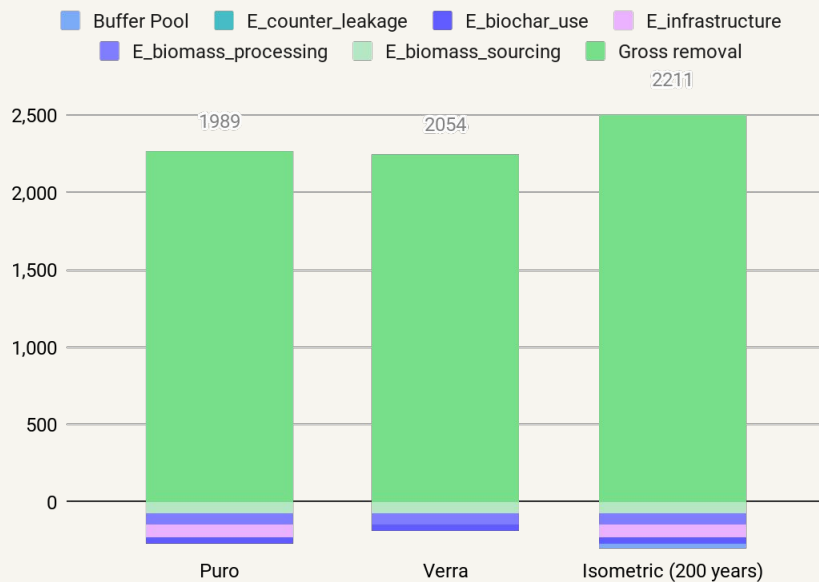
- Puro 2025 V1 and Isometric V1.2 (200-year models) both use decay models based on the H/Corg ratio and soil temperature, based on [Woolf et al. \(2021\)](#) with different interpretation and advances. For this project, the resulting permanence factor is 0.90 under Isometric V1.2 and 0.81 under Puro 2025 V1.
- Verra uses a simpler method based on pyrolysis temperature. For medium-temperature pyrolysis (450–600°C) (i.e., this project's case), the permanence factor is 0.80; for high-temperature systems (>600°C), it will be 0.89.

Credit Volume Estimation

Scenario Analysis Indicates Up to 8.5% Over-Crediting Risk Under Current Assumptions

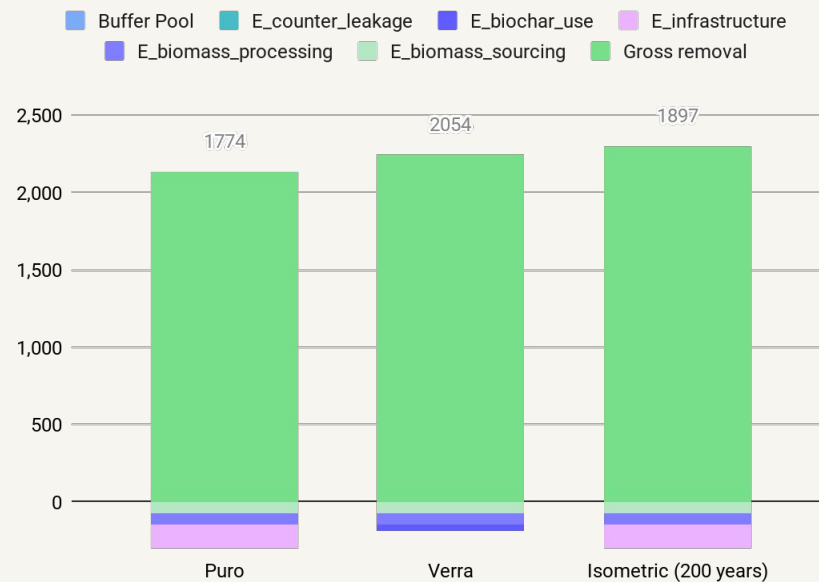
Target Scenario

Annual Credit Issuance ($H/C=0.200$; 20 year amortisation)



Conservative Scenario

Annual Credit Issuance (Year 1-10) ($H/C=0.315^*$; 10 year amortisation)



* The H/C benchmark is sourced from Sylvera's internal database, based on wood-based feedstocks and high-tech pyrolysis systems. In general, lower H/C ratios can be achieved by increasing pyrolysis temperatures (typically above 500°C), extending residence time, and using lignin-rich, low-moisture feedstocks.

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Methodology Assessment Comparison Summary (1/2)

Integrity pillar	Quality indicators	Rationale	Isometric	Puro	Verra
Carbon accounting	Gross removals quantification	Gross removals must be measured transparently using validated models to avoid inflated crediting	●	●	●
	Feedstock counterfactual	Eligible biomass must be assessed against its alternate fate to prevent overstating removals	●	●	●
	Baseline credibility	Conservative baselines are required to ensure only real, additional removals are credited	●	●	●
	Project emissions quantification	All project-related emissions must be quantified and deducted to calculate net removals	●	●	●
	LCA quality	Robust cradle-to-grave LCA boundaries and allocation choices determine accuracy of removals	●	●	●
	Data quality	Indirect emissions from market displacement must be included to avoid over-crediting	●	●	●
	Leakage	Transparent, auditable and high-quality data underpin the reliability of carbon accounting	●	●	●
Additionality	Eligibility tests	Methodologies must set strict criteria on feedstock, biochar quality, and use to prevent crediting unsustainable or short-lived removals	●	●	●
	Financial additionality tests	Carbon revenues must demonstrably close the economic viability gap for the project	●	●	●
	Policy and regulatory surplus	Projects required by law or already subsidised cannot be considered additional	●	●	●
	Common practice	Widespread adoption of similar projects undermines the case for additionality	●	●	●
	Double counting	Clear ownership and registry controls are needed to prevent double issuance or claims	●	●	●
	Monitoring	Ongoing checks ensure additionality remains valid as market or policy conditions evolve	●	●	●

Methodology Assessment Comparison Summary (2/2)

Integrity pillar	Quality indicators	Rationale	Isometric	Puro	Verra
Permanence	Risks Assessment	Comprehensive risk assessments must cover technical, natural, and anthropogenic reversal pathways			
	Monitoring	Continuous monitoring and timely reporting of reversals are essential safeguards			
Safeguarding and Co-Benefits	Community safeguarding	Projects must ensure no social harm and include robust engagement processes			
	Environmental safeguarding	Activities should follow strict "do no net harm" principles for ecosystems and biodiversity			
	Sustainable Development Goals (SDGs)	Projects should demonstrate credible alignment with relevant SDG contributions			

Carbon accounting

Overview of Sylvera’s approach to assessing carbon accounting

Pillar	Parameter	Why it's important
Carbon accounting	Gross removal quantification	Gross carbon removal represents the total carbon sequestered in biochar before deductions. Requires accurate measurement of biochar yield, fixed carbon content, and long-term stability using validated decay or durability models. Overestimation can inflate total credited removals.
	Feedstock counterfactual	The counterfactual fate of biomass should be assessed to ensure only waste, residues, or invasive species are used, and to prevent reliance on living forests or standing carbon stocks. This avoids perverse incentives to harvest trees, protects existing carbon sinks, and ensures removals are not overstated through the use of biomass that would otherwise remain sequestered in natural ecosystems.
	Baseline credibility	A credible baseline ensures that only additional carbon removals are credited. The project should use a conservative baseline estimation method that mitigates the risk of inflated baseline emissions. The project should also distinguish between avoided and removal emissions.
	Project emissions quantification	To calculate a net removal value, the GHG emissions associated with the project must be estimated and subtracted. This is done using Life Cycle Assessment (LCA), and significantly impacts the net removal value.

Carbon accounting

Overview of Sylvera’s approach to assessing carbon accounting

Pillar	Parameter	Why it's important
Carbon accounting	LCA quality	The quality of modelling decisions used in the LCA significantly affect the outcome of the project's LCA. This gives a view on whether the project is likely to have overestimated or accurately estimated their project emissions, and therefore their net removals value.
	Data quality	The values used to estimate the carbon pools and emissions, as well as their sources, determine the confidence level of the net benefits claimed by projects. For accurate estimates, methodologies should recommend that primary data is used, and high-quality secondary data when the former is unavailable. Primary data come from the actual process, including direct emissions measurements or materials used, and should be supported by documentation. Secondary data usually consist of average industry emission factors; the best sources include the Ecolnvent database, the GaBi database, official organisations and other reviewed sources. The data should be geographically, technologically, and temporally representative of the process in scope. The most conservative approach involves using reputable, peer-reviewed models with their sources disclosed, tailored to the specific needs of each project.
	Leakage	Project activities may incur emissions from displaced activities, market shifts, or land-use change. These emissions should be assessed and mitigated to avoid overstating net removals.

Carbon accounting findings – Isometric



Overall, Isometric’s carbon accounting risk is low and generally aligns with best practice; some risk remains due to project-level discretion in emissions accounting.

RAG

Parameters	Isometric Biochar Protocol v1.2 Summary	Strengths and risks	RAG
Gross removal quantification	For application in agricultural soils, gross removals are adjusted by a durability fraction, with two durability tiers: a 200-year decay model using soil temperature and H/C_{org} with fixed regression parameters based on Woolf et al. (2021) that employs a lower-bound durability curve (17th percentile) to ensure persistence is underestimated; and a 1,000-year option using random reflectance (R_0) relative to inertinite to derive a fraction of inert biochar. Each production batch can only be verified under one durability tier. Carbon sampling frequency and outlier treatment are defined to maintain representativeness across production batches. Durable fraction determined using R_0 for other end-use pathways.	Strengths: Responsive to evolving biochar science through pathway-specific quantification rather than a single decay model. The reflectance-based 1,000-year tier in soils aligns with emerging geologic inert benchmarks and introduces scientific rigor. Per-batch quantification supports traceability and avoids double counting between durability tiers. Risks: Inherent risks exist with gross removal calculations across registries due to rapidly evolving research landscape for biochar decay rates which leads to variability. Despite its scientific backing, the methods employed under the 200-year decay model yield higher permanence factors compared to other registries (Puro, CSI). The optional dual-pathway system introduces comparability challenges across projects.	
Feedstock counterfactual	Biomass must meet Isometric Biomass Sourcing Criteria which provides evidence-based framework for determining whether biomass feedstocks qualify as additional and non-depletive of existing carbon sinks. Eligible categories include agricultural residues, forestry thinnings, industrial residues, municipal wastes, and invasive species. Forest residues must meet strict sustainability criteria verified through certification. Counterfactual storage assessments are required every ten years.	Strengths: The sustainability criteria, counterfactual storage calculation, and market-leakage components together prevent removals from being overstated by ensuring that only residues, wastes, and invasive species are eligible. Strong documentation and traceability requirements enforce feedstock eligibility. Stringent restrictions on non-waste biomass sources. Decadal reassessment of counterfactuals maintains relevance as regional biomass markets and management practices evolve.	

Carbon accounting findings – Isometric



Overall, Isometric’s carbon accounting risk is low and generally aligns with best practice; some risk remains due to project-level discretion in emissions accounting.

RAG

Parameters	Isometric Biochar Protocol v1.2 Summary	Strengths and risks	RAG
Baseline credibility	CO ₂ e _{counterfactual} parameter encompasses carbon that would have been durably stored in the baseline scenario. Must be quantified using the Biomass Feedstock Accounting Module to ensure only additional removals are credited.	Strengths: Conservative non-zero baselines enforced in scenarios where carbon storage occurs without the project, reducing over-crediting risk. No allowance for baselines less than 0 reducing risk of baseline inflation.	
Project emissions quantification	GHG Statement covering all sources, sinks, and reservoirs (SSRs) within the system boundary with equations defined under emissions accounting sub-modules. Direct measurement of pyrolysis gases is required through two options: direct continuous measurement (primary and preferred) using gas flow rate and composition or regular emissions testing where continuous measurement is not feasible. The GHG Statement must document emission factors, data sources, and justifications for any exclusions.	Strengths: Mandatory GHG Statement ensures full-boundary life-cycle reporting and traceability across all emission sources. Defined equations and boundary tables promote procedural consistency, while direct testing of process gases improves accuracy. Nearly all material sources must be included, and any omission must be transparently justified and auditor-approved. The Isometric Certify platform provides a guided digital workflow for building LCAs and generating GHG Statements, improving standardisation, transparency, and validation readiness across projects.	
LCA quality	Cradle-to-grave system boundaries prescribed. The functional unit is one tonne of CO ₂ e removed and stored for the reporting period. Projects must allocate emissions between co-products or by-products according to their functional use or economic value, with justification provided in the GHG Statement. Only net removals are credited; avoided emissions are not eligible for crediting. Exclusions of SSRs are permitted only with documented justification.	Strengths: Mandatory cradle-to-grave boundary and prohibition of avoided emissions. Certify tool provides a structured template and embedded guidance for defining functional units, system boundaries, and allocation methods, reducing methodological variability and improving replicability. Risks: Allowance of allocation may enable selective minimization of embodied emissions or the unreasonable shifting of burdens to co-products, leading to under-accounting.	

Carbon accounting findings – Isometric

ISOMETRIC

Overall, Isometric’s carbon accounting risk is low and generally aligns with best practice; some risk remains due to project-level discretion in emissions accounting.

RAG

Parameters	Isometric Biochar Protocol v1.2 Summary	Strengths and risks	RAG
Data quality	Requires documentation of all measurement data, emission factors, and monitoring records in the GHG Statement. Direct emissions testing must use calibrated instruments and maintain “Required Records” including permits, compliance evidence, and logs of measurement accuracy and frequency. Sampling protocols for biochar carbon content specify representativeness, outlier treatment, and frequency.	Strengths: Comprehensive record-keeping and calibration requirements ensure traceability and auditability of key measurements. Batch-level sampling, outlier handling, and accuracy verification support consistent carbon quantification across production. Risks: Preferred data hierarchy between primary and secondary sources undefined. No guidance provided on suitable emission factor sources, which introduces variability in data quality across projects. Thus emission factor selection and data modelling are developer-led which risks non-representativeness of sources.	
Leakage	Leakage must be identified, mitigated, and accounted for in total project emissions. The Biomass Feedstock Accounting Module requires assessment of displacement risks and market impacts for each feedstock type at minimum. Qualitative justification of low leakage must be provided in the PDD, and any material residual leakage must be quantified and deducted from net removals.	Strengths: Explicit requirement to identify and include leakage within project emissions and stringent requirements to quantify potential leakage impacts ensure boundary completeness and discourages over-crediting.	

Carbon accounting findings – Puro puro·earth

Similarly, Puro’s carbon accounting risk is low and broadly aligns with best practice; however, some risk remains due to project-level discretion that could lead to undercounted emissions.

RAG

Parameters	Puro Biochar 2025 V1 Summary	Strengths and risks	RAG
Gross removal quantification	Uses a decay-based model derived from Woolf et al. 2021 and inclusive of more recent scientific research as of 2025 (Azzi et al., 2024) (Li et al., 2024) (Sanei et al., 2025) and some VCM considerations. Permanence factor determined as a function of H/C_{org} with regression parameters defined for varying soil temperatures with an embedded 80% confidence interval lower bound. Adopted less conservative power model in 2025 over a previous multi-pool exponential decay model as it is more representative of biochar decay rate.	Strengths: Responsive to scientific advances. Conservative persistence model using lower 80% confidence interval. Explicit temperature dataset with minimum 7 °C. H/C_{org} threshold enforces recalcitrance with increasing carbonisation. Per-batch quantification supports traceability. Risks: Inherent risks exist with gross removal calculations across registries due to rapidly evolving research landscape for biochar decay rates. While the new model is likely more representative of biochar persistence, it is a less conservative decay model than previous and in comparison to other methodologies. However, this risk is mitigated by an extended time horizon and the lower bound confidence interval. The H/C ratio alone may be insufficient for determining biochar stability as it represents bulk elemental composition that can mask carbonization heterogeneity allowing crediting of 200+ year durability on carbon fractions more susceptible to degradation on a decadal time-scale. While it may not affect credit quantification, project’s may provide supplementary characterization data (random reflectance) to increase confidence.	
Feedstock counterfactual	Biomass must meet Puro Biomass Sourcing Criteria with defined allowed categories and exclusions. Invasive species allowed. Food crops and mixed municipal solid waste excluded. Forest biomass are allowed only if strict sustainability criteria are met. Batch records and traceability required for feedstock eligibility.	Strengths: Clear category lists and exclusions protect standing carbon stocks. Invasive species pathway supported. Strong documentation and traceability requirements enforce feedstock eligibility. Stringent restrictions on non-waste biomass sources.	

Carbon accounting findings – Puro

Similarly, Puro’s carbon accounting risk is low and broadly aligns with best practice; however, some risk remains due to project-level discretion that could lead to undercounted emissions.

RAG 

Parameters	Puro Biochar 2025 V1 Summary	Strengths and risks	RAG
Baseline credibility	$C_{baseline}$ is defined as CO₂e that would have been stored in the baseline scenario and must be calculated for retrofit and charcoal repurpose facilities as defined by the methodology. Projects under these baseline scenarios must calculate carbon stored from persistent char in the baseline scenario to remove from overall CORC calculation.	Strengths: Conservative non-zero baselines enforced in scenarios where carbon storage occurs without the project, reducing over-crediting risk in retrofit and charcoal repurpose contexts. No allowance for baselines less than 0.	
Project emissions quantification	LCA required following general principles defined in ISO-14040/44 and the ISO-14064. Digital LCA model and LCA model description required and validated at Production Facility Audit.	Strengths: Mandatory LCA modelling demonstrates procedural rigour. LCA template model provided with allowance for equivalent with appropriate level of transparency, detail and verifiability. Risks: Materiality thresholds for potentially negligible emissions are only qualitatively defined allowing exclusion of emission sources without a numeric threshold. This increases reliance on auditor judgement and carries a risk of undercounting small but cumulative emission sources.	
LCA quality	Cradle-to-grave system boundaries prescribed. Functional unit required to cover production and end-use of biochar. Only removals are credited and avoided emissions are explicitly excluded. Allocation of emissions between co-products allowed on evidence of meaningful use.	Strengths: Mandatory cradle-to-grave boundary and prohibition of avoided emissions. Guidance provided with LCA template. Stage-specific allocation rules are clearly defined, offering transparency and uniformity across project types. Embodied emissions associated with infrastructure and equipment are amortized evenly over the shorter of the first crediting period or the assumed lifetime of the production facility, with the option for upfront accounting during the first monitoring period when requested by investors or buyers. Risks: Allowance of allocation may enable selective minimization of embodied emissions or the unreasonable shifting of burdens to co-products, leading to under-accounting.	

Carbon accounting findings – Puro

Similarly, Puro’s carbon accounting risk is low and broadly aligns with best practice; however, some risk remains due to project-level discretion that could lead to undercounted emissions.

RAG 

Parameters	Puro Biochar 2025 V1 Summary	Strengths and risks	RAG
Data quality	Mandates that an LCA Model Description be provided, documenting data sources, emission factors, assumptions, and monitoring methods, all of which must be auditable and transparently justified. Combined uncertainty of all measured parameter must be quantified guided by ISO/IEC Guide 98-3 or EURACHEM-CITAC Guide CG 4 .	Strengths: Insistence on transparent documentation, uncertainty quantification, and audit-ready data records supports traceability and accountability. Time-stamped monitoring records also facilitates ex-post audit verification and reduces risks of retrospective manipulation. Risks: Preferred data hierarchy between primary and secondary sources undefined.	
Leakage	Leakage to be addressed with actions to avoid or mitigate as appropriate. Potential sources must first be identified, then avoided or mitigated where feasible, and any residual leakage must be quantified and incorporated into the CORC calculation. Recognises both ecological leakage (e.g., nutrient displacement, soil carbon disturbance) and market or activity-shifting leakage (e.g., diverted biomass demand, indirect land-use change).	Strengths: Comprehensive acknowledgement of risks beyond the project boundaries. Inclusion of an indirect land-use change (iLUC) screen strengthens integrity where biochar projects rely on feedstocks that may influence agricultural or forestry markets. Risks: Complex displacement effects, particularly in biomass markets or land systems, risk being underestimated or overlooked due to a lack of prescribed quantification methods but likely to be insignificant or qualitatively addressed.	

Carbon accounting findings – Verra

Verra’s approach to carbon accounting introduces high risks to integrity due allowances for non-conservative defaults and assumptions and a lack of granular decay modelling.

RAG 

Parameters	VCS VM0044 v1.2 Summary	Strengths and risks	RAG
Gross removal quantification	Derived from measures mass of stable organic carbon in biochar and adjusted by 100-year permanence factor according to end-use pathway and pyrolysis temperature. The permanence adjust factor ($PR_{de,k}$) applied is a default value drawn from IPCC (2019) and Woolf et al. (2021), with three options depending on pyrolysis temperature range ($>600^{\circ}C$, $450 - 600^{\circ}C$, or $350 - 450^{\circ}C$) for soil and non-soil applications. Projects may suggest lower and literature-backed $PR_{de,k}$ for non-soil. For lower-technology systems where production temperature is uncertain, the method allows use of conservative default organic carbon content values from literature according to feedstock and production-type (pyrolysis or gasification) and applies a default $PR_{de,k}$ of 0.56.	Strengths: Provides a transparent and replicable framework for quantifying stable carbon in biochar, with clear equations and parameter definitions that minimize interpretive ambiguity. The inclusion of conservative defaults for lower-tech production ensures inclusivity of small producers while maintaining a relatively conservative bias in the absence of analytical testing. Risks: Permanence modelling is less granular than emerging approaches. Tying long-term stability to a weak proxy like the pyrolysis temperatures significantly risks crediting non-durable fractions of biochar. This method overlooks other critical parameters such as H/C_{org} or reflectance increasingly used in more recent standards. Default permanence factors are derived from limited datasets, reducing confidence that credited removals correspond to empirically supported persistence.	
Feedstock counterfactual	Eligible feedstock defined as biomass residues, wastes, or by-products that would otherwise be left to decay, burned without energy recovery, or disposed. Mandates that projects provide documentary evidence and attestations demonstrating that the biomass meets this waste-only criterion, referencing activity over the preceding five years. For forestry residues, the methodology requires sourcing from sustainably managed forests verified through certification schemes such as FSC, PEFC or equivalent.	Strengths: Clear category definitions, exclusions, and multi-year documentation requirements create strong procedural safeguards around feedstock eligibility. The prohibition of purpose-grown biomass and certification requirement for forest residues enhance sustainability. Risks: For non-forest biomass, the methodology relies primarily on attestations and qualitative documentation to confirm waste status, without independent certification. As the methodology’s assumptions on additionality, baseline, and leakage depend on the feedstock being genuine waste, any misclassification of non-forest materials could materially affect credited removals.	

Carbon accounting findings – Verra

Verra’s approach to carbon accounting introduces high risks to integrity due allowances for non-conservative defaults and assumptions and a lack of granular decay modelling.

RAG 

Parameters	VCS VM0044 v1.2 Summary	Strengths and risks	RAG
Baseline credibility	The baseline is set to a default value of zero, representing no removals or avoided emissions absent of the project activities.	Strengths: No allowance for baselines less than 0 reducing risk of baseline inflation. Risks: No acknowledgement of potential positive non-zero baselines where carbon may have been durably stored in the baseline scenario or feedstock counterfactual absent of the project activities.	
Project emissions quantification	Project emissions include pre-treatment, production, and application stages with equations defined for each. High-technology systems may assume negligible production emissions (=0) while low-tech kilns must estimate CH ₄ using literature-based factors. Energy consumed during feedstock preparation and auxiliary operations is quantified using CDM methodological tools for fossil-fuel and electricity use.	Strengths: Structured equations and standardized CDM tools ensure consistent estimation across projects. Risks: The absence of a verifiable, ISO-aligned LCA model significantly weakens transparency and completeness. Zero-emission defaults for high-tech systems are unjustified and risk systematic underestimation of real process emissions.	
LCA quality	Boundaries encompass feedstock sourcing, biochar production, and application stages. Emission sources within these stages are quantified through defined equations and default parameters, rather than through a formalized LCA model. The methodology does not prescribe or require the submission of a verified LCA.	Risks: Absence of a formalized LCA requirement means that project-specific emission pathways and indirect effects are not comprehensively captured. Reliance on generalized defaults and predefined boundaries limits the representation of site-specific variability and process efficiency. Consequently, results may diverge from those produced under methodologies requiring empirical or ISO-aligned LCA modeling	

Carbon accounting findings – Verra

Verra’s approach to carbon accounting introduces high risks to integrity due allowances for non-conservative defaults and assumptions and a lack of granular decay modelling.

RAG 

Parameters	Verra VM0044 v1.2 Summary	Strengths and risks	RAG
Data quality	Specifies documentation of all measurement data, laboratory analyses, and monitoring parameters. Laboratory testing must follow recognized standards such as International Biochar Initiative (IBI) or European Biochar Certificate (EBC), using calibrated instruments and maintaining QA/QC records. Required records include calibration logs, sampling documentation, and geolocation of all application sites, with traceability maintained through batch-level records across the production and application chain.	Strengths: Defined monitoring parameters and the use of accredited laboratory standards ensure procedural consistency and traceability. Batch-level documentation, calibration requirements, and geospatial traceability enhance auditability and data verification. Risks: Preferred data hierarchy between primary and secondary sources undefined. Quantitative uncertainty analysis is not required, and data validation relies primarily on procedural compliance rather than statistically characterized accuracy, potentially limiting comparability with ISO-aligned data management approaches.	
Leakage	Only leakage emissions from transport are deemed material. Transport emissions can be assumed zero where distances are <200km (round trip). Activity-shifting and biomass-diversion leakage are set to zero on the basis that all feedstocks must qualify as waste and therefore do not displace existing uses.	Risks: Prescription of zero-assumption for activity shifting and biomass diversion of leakage severely risks not representing emissions driven outside system boundaries by project activities. Assumption relies on waste-classification of feedstock which presents risks of integrity due to non-stringent sourcing criteria as assessed earlier. The 200 km threshold may also exclude cumulative short-distance transport emissions from dispersed collection and application sites.	

Additionality

Overview of Sylvera’s approach to assessing additionality

Pillar	Parameter Assessed	Why it’s important
Additionality	Eligibility tests	The minimum standards a methodology has in place to establish whether a project can enroll in the methodology are key for setting expectations of additionality. Strong eligibility standards prevent crediting of projects using unsustainable feedstocks, low-quality biochar, or short-lived applications that would not deliver durable removals.
	Financial additionality tests	Methodologies should require projects to demonstrate that carbon finance is necessary for viability. This can be through cost, investment, or barrier analyses, including disclosure of subsidies and external revenue streams, to ensure removals would not occur without credit revenues.
	Policy and regulatory surplus	Methodologies should require projects to show that their activities go beyond existing laws, regulations, or mandatory incentives. Demonstrating surplus ensures credited removals are not already required or incentivized by public policy.
	Common practice	Methodologies should include a check to make sure similar interventions are not already commonly practiced in the project’s specific geographic or socio-economic context. If the activity is already widespread - especially without carbon finance - the project may not be additional.

Additionality

Overview of Sylvera’s approach to assessing additionality

Pillar	Parameter Assessed	Why it's important
Additionality	Double counting risk	Without robust checks within methodologies, projects could retire credits from the same carbon removals across multiple issuers or programs. To prevent risk of over-crediting, methodologies should include preventative measures to avoid double issuance, double use, and double claiming.
	Monitoring	A robust methodology should include requirements for monitoring periods, with more frequent assessments required when regulatory or legal requirements change or when project financials suggest that carbon financing is no longer necessary, appears reasonable. Enhanced robustness in monitoring should also account for changes in political transitions and evolving common practices.

Additionality findings – Isometric



Although Isometric includes extensive additionality tests limiting risks to additionality, some assessment approaches are less robust which may pose some risk.

RAG

Parameters	Isometric Biochar Protocol v1.2 Summary	Strengths and risks	RAG
Eligibility tests	Eligibility determined through Biochar Protocol and Feedstock Module. Forestry feedstocks must hold certification under schemes such as FSC, PEFC, SFI or equivalent. All feedstocks must meet sustainability, counterfactual storage, and market leakage criteria, with avoided emissions excluded and only durable carbon storage uses permitted. Biochar chemical requirements defined for soil applications as H/C_{org} ratio <0.5 or Random Reflectance >2%.	Strengths: Mandatory third-party certification for forestry residues establishes strong sustainability assurance. Dedicated feedstock criteria and exclusion of avoided emissions maintain a strict carbon removal boundary. Conservative biochar quality and stability thresholds are defined. Risks: Certification coverage is limited to forestry residues, with agricultural and industrial residues often relying on supplier or local authority documentation.	
Financial additionality tests	Full financial additionality test only required if removal credits are not the only revenue source. In which case, projects must provide an IRR analysis without carbon finance showing ≤0% IRR or IRR below the required hurdle rate. Analysis must include subsidies/tax incentives and be completed over a 10-year horizon. A robust sensitivity analysis is also required.	Strengths: Mandatory rigorous investment analysis as well as public subsidy disclosure ensure transparency and strengthen confidence, particularly with the inclusion of sensitivity analysis requirement. Risks: In the case that the project only earns revenue from carbon, there are very limited financial additionality tests required by Isometric. Although financial additionality is implied in this scenario, a uniform requirement for investment analysis across all projects would still improve consistency and strengthen confidence in financial additionality. Additionally, Isometric allows high-risk projects to justify a higher IRR as additional, which may be misleading.	

Additionality findings – Isometric ISOMETRIC

Although Isometric includes extensive additionality tests limiting risks to additionality, some assessment approaches are less robust which may pose some risk.

RAG

Parameters	Isometric Biochar Protocol v1.2 Summary	Strengths and risks	RAG
Policy and regulatory surplus	Projects must demonstrate that the project is not required by existing laws, regulations, policies, or other binding obligations. If the Project is legally required but removals exceed the legal requirement, the excess removals can be considered additional.	Strengths: The regulatory surplus test is mandatory. Adherence to Isometric's requirements are likely to reduce risks that projects would already be required by national policies, which strengthens the case of financial additionality. Projects are also required by Isometric to include existing government subsidies or tax incentives as a revenue stream. Risks: Allowing excess removals to be credited when the project activity is mandated by law can introduce additionality risks, as it can be argued that these removals could have happened in the absence of carbon finance.	
Common practice	The common practice test only becomes a requirement once the technology reaches a threshold of operational maturity determined by a Technology Readiness Level (TRL) of 8 or 9. At which point, projects must assess the market penetration of biochar production and use within their region and sector to show that their activity remains uncommon without carbon finance.	Risks: Isometric defines Biochar TRL as 6-7, estimated by IPCC, which introduces high risks to integrity. TRL scores can become outdated as technologies evolve rapidly, and they do not necessarily reflect current real-world market penetration. Also, projects using commercially widespread but technically mature technologies could still qualify as additional.	
Double counting	Double counting addressed through strict ownership requirements and contractual chain-of-title. A single Project Proponent must hold exclusive rights to all carbon removals across the entire biochar value chain. The registry enforces unique issuance and retirement of credits to ensure removals cannot be claimed elsewhere, and ownership evidence is verified at validation.	Strengths: The requirement for a single accountable Project Proponent and documented ownership across every stage of the value chain reduces risks of overlapping claims between suppliers, processors, and end-users. Registry-level issuance and retirement controls further ensure each tonne of removal is credited only once, preventing duplication across carbon markets or corporate claims.	

Additionality findings – Isometric



Although Isometric includes extensive additionality tests limiting risks to additionality, some assessment approaches are less robust which may pose some risk.

RAG

Parameters	Isometric Biochar Protocol v1.2 Summary	Strengths and risks	RAG
Monitoring	Additionality must be reassessed at verification if project conditions materially change (e.g., financial viability or regulatory mandates). Verification occurs at least every five years, this creates an implicit maximum interval for reviewing additionality.	Strengths: Additionality is subject to reassessment at each verification if material conditions change, such as new regulatory mandates or improved financial viability without carbon finance. This event-based review mechanism helps ensure continued eligibility over time. Risks: Reliance on verification-triggered reassessment means structural market shifts, such as rapid scaling of biochar or indirect policy incentives, may not be captured unless explicitly disclosed by the project. With common practice tests deferred until TRL maturity, there is limited protection against emerging market normalization.	

Additionality findings – Puro

Although Puro includes extensive additionality tests limiting risks to additionality, some assessment approaches are less robust which may pose some risk.

RAG



Parameters	Puro Biochar 2025 V1 Summary	Strengths and risks	RAG
Eligibility tests	Puro's eligibility framework sets clear sustainability and quality thresholds. Biochar projects must use sustainably sourced biomass (e.g., FSC, PEFC, RSB, ISCC certified), demonstrate controlled pyrolysis ($H/C_{org} < 0.7$), and ensure that credited applications are for stable, long-term storage. Avoided emissions are excluded from crediting.	Strengths: Requires certified or equivalent proof of sustainable biomass sourcing through schemes such as FSC, PEFC, RSB, or ISCC. These certifications ensure traceable, environmentally responsible feedstocks, minimizing risks of deforestation, land-use change, and biodiversity loss. Puro's biochar quality and use requirements are well-defined, mandating controlled pyrolysis and restricting credits to stable soil or durable material applications. Combustion is prohibited. Annual laboratory testing, end-use documentation, and independent auditor verification provide high assurance of traceability. The methodology only credits carbon removals, explicitly excluding reduced or avoided emissions. As baselines must be set at zero, the uncertainty and double-counting risk associated with avoided emissions is minimized. Risks: Allowing projects to demonstrate biomass sustainability through "equivalent" documentation instead of formal certification can create inconsistency in assurance across projects. If such claims are not rigorously verified by auditors, there is a small risk that non-certified feedstocks or practices may not fully meet environmental standards. It also allows durable non-soil uses, but cascading end-use pathways may slightly weaken eligibility tests.	
Financial additionality tests	Projects must perform a financial additionality test, choosing from simple cost, investment, or barrier analyses depending on project profile. The methodology requires reporting of public subsidies and allows use of CDM tools for investment justification.	Strengths: Mandatory financial analysis as well as public subsidy disclosure ensure transparency and strengthen confidence. Risks: Test robustness varies as Puro allows flexibility in how projects demonstrate financial additionality, permitting either a cost analysis or an investment analysis depending on the project's financial profile e.g., whether the project relies solely on carbon credits for income or has minimal CapEx compared to OpEx. A uniform requirement for investment analysis across all projects would improve consistency and strengthen confidence in financial additionality. Also, expanding public subsidy disclosure provision to cover all sources of external financing would reduce risks of unaccounted-for revenue streams.	

Additionality findings – Puro

Although Puro includes extensive additionality tests limiting risks to additionality, some assessment approaches are less robust which may pose some risk.

RAG

Parameters	Puro Biochar 2025 V1 Summary	Strengths and risks	RAG
Policy and regulatory surplus	Projects must demonstrate compliance with applicable laws and confirm that biochar production and use are not legally required or publicly incentivized. Projects must also disclose public subsidies and policy supports, ensuring activities are beyond regulatory mandates.	Strengths: The methodology requirements are likely to reduce risks that projects would already be required by national policies and strengthen additionality assurance.	
Common practice	The common practice test only becomes a requirement once the technology reaches a threshold of operational maturity determined by a Technology Readiness Level (TRL) of 8 or 9. At which point, projects must assess the market penetration of biochar production and use within their region and sector to show that their activity remains uncommon without carbon finance.	Risks: Puro defines Biochar TRL as 6-7, estimated by IPCC, which introduces high risks to integrity. TRL scores can become outdated as technologies evolve rapidly, and they do not necessarily reflect current real-world market penetration. Also, projects using commercially widespread but technically mature technologies could still qualify as additional.	
Double counting	The Puro Standard issues one CORC per verified tonne of CO ₂ removed, tracked through a centralized registry. Each credit has a unique identifier and full chain-of-custody, preventing issuance, transfer, or retirement duplication. Projects must also confirm they are not simultaneously registered under another crediting program for the same period or activity.	Strengths: The registry's transparency and ownership checks substantially reduce double-counting risks. Mandatory documentation of credit ownership and end-use traceability ensures removals cannot be credited twice within or outside Puro.	

Additionality findings – Puro

Although Puro includes extensive additionality tests limiting risks to additionality, some assessment approaches are less robust which may pose some risk.

RAG 

Parameters	Puro Biochar 2025 V1 Summary	Strengths and risks	RAG
Monitoring	Puro Standard conducts biennial reviews of its Additionality Assessment Requirements, but individual projects are not automatically re-assessed with each update. Projects are required to re-validate additionality with each production facility audit i.e., at the renewal of the crediting period, defined as every 10 years, to the latest standard.	Risks: This is not sufficiently robust and poses a risk of failing to capture potential policy and regulatory changes throughout the project period. Monitoring requirements could be enhanced by incorporating more frequent assessments of the relevant regulatory landscape or opting-in to re-assess against newer standards during the crediting period.	

Additionality findings – Verra

While Verra includes extensive tests to limit risks to additionality, some assessment approaches are less robust which may pose some risk.

RAG

Parameters	Verra VM0044 v1.2 Summary	Strengths and risks	RAG
Eligibility tests	Limits eligibility to new biochar facilities using waste biomass only, excluding purpose-grown feedstocks. Feedstocks must not cause deforestation or degradation, and forestry residues must come from sustainably managed sources, evidenced through examples such as approved management plans or certification under FSC, PEFC, or equivalent schemes. Biochar must meet IBI/EBC standards, achieve $H/C_{org} \leq 0.7$, and be used within one year of production in soil or durable non-soil applications.	Strengths: Sets a clear environmental boundary by requiring waste-only feedstocks, excluding avoided-emission activities, and referencing recognized sustainability and quality standards. Risks: FSC, PEFC, and other schemes are examples rather than mandatory, assurance quality depends on verifier judgment and local documentation standards. Reliance on self-declared or “equivalent” evidence could create inconsistent verification of sustainability.	
Financial additionality tests	Assessment required through VCS Tool to analyse additionality (Step 3 in VT0008). Financial additionality can be evaluated using either an Investment Comparison Analysis (comparing financial metrics ⁽¹⁾ against counterfactual scenarios to prove carbon credits make the project more financially attractive) or a Benchmark Analysis (comparing IRR against a benchmark indicator to prove project activity does not meet this benchmark in the absence of carbon credits).	Strengths: Mandatory financial analysis as well as public subsidy disclosure promote transparency and strengthen confidence. These requirements mean that investment barriers in the absence of carbon credit funding can be adequately captured. Projects are expected to factor possible future costs of carbon into their equations, indicating comprehensive analysis. Risks: Test robustness varies as Verra allows flexibility in how projects demonstrate financial additionality, permitting either an investment comparison analysis or a benchmark analysis. A uniform requirement for investment analysis across all projects would improve consistency and strengthen confidence in financial additionality. The project does not prohibit additional revenues, but it does mandate the disclosure of these in its analysis, reducing possible impact.	

(1) Financial metric could be either IRR, NPV, cost-benefit ratio, levelised cost

Additionality findings – Verra

While Verra includes extensive tests to limit risks to additionality, some assessment approaches are less robust which may pose some risk.

RAG 

Parameters	Verra VM0044 v1.2 Summary	Strengths and risks	RAG
Policy and regulatory surplus	Demonstration of regulatory surplus required. Projects must confirm that biochar production and use are not mandated by any law, regulation, or enforced policy.	Strengths: The regulatory surplus test is mandatory under Verra's framework, reducing the risk that credited activities are already required by national or regional policies. This strengthens overall additionality assurance. Financial analysis under VT0008 requires transparent disclosure of any government subsidies or concessional finance, improving credibility and reducing the risk of overstating dependence on carbon credit revenues.	
Common practice	Addressed through VM0044's activity-method positive list, which deems projects that meet all eligibility requirements as additional (e.g. new pyrolysis facility for waste biomass) and represent a low market share. This approach substitutes for a separate market-penetration test.	Strengths: Embedding the positive list within eligibility criteria provides a clear, low-discretion framework and simplifies validation while maintaining consistency across projects. Risks: The positive-list approach is static and not automatically updated. If biochar adoption expands or economic conditions change, activities could remain automatically additional even after becoming common practice. VM0044 includes no requirement for periodic reassessment of market penetration or technological diffusion, creating a risk that the additionality justification may become outdated over time.	
Double counting	The methodology prohibits projects from claiming other environmental credits from the given activities.	Strengths: Requirements are in place to prevent double issuance, double claiming, and double use according to the VCS Standard v4.7 .	

Additionality findings – Verra

While Verra includes extensive tests to limit risks to additionality, some assessment approaches are less robust which may pose some risk.

RAG 

Parameters	Verra VM0044 v1.2 Summary	Strengths and risks	RAG
Monitoring	The methodology does not provide guidance on monitoring additionality and makes no reference to required updates in project additionality assessments throughout its lifetime.	Risks: Monitoring of additionality is left at the developer's discretion, which is not a conservative approach. Accounting for changes through project crediting that can affect the additionality of the project needs to be captured to ensure the integrity and effectiveness of carbon credits. Examples of such changes include political transitions, adjustments in policy and regulatory frameworks, and evolving common practices.	

Permanence

Overview of Sylvera’s approach to assessing permanence

Pillar	Parameter Assessed	Why it’s important
Permanence	Risks Assessment	The scope of risk assessments required (internal, external, natural) and mitigation requirements can determine how comprehensively a project accounts for its potential reversal pathways. For biochar projects, this includes requirements for biochar stability and end-use, as well as safeguards against carbon stock loss before and after final application.
	Monitoring	Stringent monitoring and reporting over time are critical to verify that credited biochar remains in eligible, durable applications. Ongoing data on end-use, handling, and site conditions provides assurance that no unreported disturbances or reversals have occurred after issuance.

Permanence findings – Isometric



Isometric shows strong safeguards to assure long term durability of carbon sequestration, as well as a buffer pool contribution, with some risks arising from limited end-use monitoring requirements.

RAG

Parameters	Isometric Biochar Protocol v1.2 Summary	Strengths and risks	RAG
Risks Assessment	Biochar stability and decay risk: For agricultural soil application, projects must demonstrate an H/C_{org} ratio < 0.5, with durability quantified via either a 200-year model using soil temperature or a 1,000-year pathway using random reflectance (R_o) with an $R_o > 2\%$. The built environment pathway treats biochar embedded in concrete or asphalt as an open system, where long-term durability is modeled through exposure to acid, abrasion, heat, or demolition over a 1,000-year horizon. The low oxygen burial pathway requires projects to maintain functionally anoxic conditions using engineered site controls and continuous oxygen monitoring.	Strengths: Stringent H/C_{org} and R_o thresholds ensure higher carbonisation requirements and thus greater resistance to decompositions from an assurance of higher aromaticity. Burial pathways applies restrictive permanence safeguards with a requirement for engineered anoxia to mitigate oxidative decay. For built materials, durability is not assumed but modeled using worst-case geochemical and mechanical scenarios, with explicit deductions for abrasion and acidic degradation. Conservative modelling strengthens environmental integrity by prioritizing under-crediting over unverifiable persistence. Issuance only after verified durable use reduces diversion/combustion risk pre-issuance. A risk-based classification system for buffer pool contribution ensures that projects exhibiting higher risk are appropriately hedging against reversals. A minimum contribution of 2% is considered robust.	
	End-use application durability: Credit issuance is only at the point of verified durable storage, with pathway-specific evidence required for soils, burial, and built applications. Reversal risks are accounted for using a risk-based buffer pool contribution. The minimum buffer pool contribution for the lowest risk projects is 2%, ranging up to 20% for high risk projects. The methodology acknowledges that biochar may pass through multiple use phases but does not prescribe mechanisms to track cascading end uses.		
		Risks: The protocol does not require ongoing verification of carbon persistence, making reversals largely unobservable. Projects can mitigate these risks by adopting end-use application site monitoring such as voluntary long-term sampling or asset stewardship. Protocol does not prescribe mechanisms to track cascading end uses.	

Permanence findings – Isometric



Isometric shows strong safeguards to assure long term durability of carbon sequestration, as well as a buffer pool contribution, with some risks arising from limited end-use monitoring requirements.

RAG

Parameters	Isometric Biochar Protocol v1.2 Summary		Strengths and risks	RAG
Risks Assessment	Misuse risk: Requires demonstrated end-use and, where risk of recovery or combustion exists, mandates physical measures such as size reduction or incorporation to prevent reuse as fuel.		Strengths: Pathway-specific safeguards minimise the likelihood of intentional misuse. Biochar size control requirements in soils reduce recovery incentives, while embedding biochar in concrete, asphalt, or anoxic storage environments provides inherent protection against re-extraction or combustion. Proactive siting rules, including slope limits, water setbacks, and geotechnical assessments, reduce exposure to erosion, excavation, or natural hazard impacts. Risks: Despite preventive controls, long-term exposure to land-use change, redevelopment, or extreme weather remains unmonitored, particularly in soil applications.	
	Product handling and storage risk: Stockpiles to be covered and kept moist or mixed with compost, with a maximum storage duration of 12 months. Any losses during storage must be accounted for prior to credit issuance. Land-use change and disturbance: Land and field management practices to be assessed and reported ahead of application. A site-suitability criteria is set for agricultural applications.			
Monitoring	Requires pathway-specific evidence such as incorporation records (soils), oxygen monitoring plans (burial), or product specifications (built environment). No ongoing post-deployment monitoring of the storage environment, and permanence is assumed through conservative modelling rather than continuous observation.		Strengths: MRV ensures permanence risks are addressed before issuance, reducing reliance on long-term supervision. Risks: No inclusion post-issuance monitoring or field-based verification of stored carbon. Reliance on modelling means any disruption, land-use change, or material disturbance could go undetected unless voluntarily disclosed or uncovered during verification.	

Permanence findings – Puro puro · earth

Puro demonstrates strong safeguards to assure long term durability of carbon sequestration, with some risks arising from limited end-use monitoring requirements.

RAG

Parameters	Puro Biochar 2025 V1 Summary	Strengths and risks	RAG
Risks Assessment	Biochar stability and decay risk: Re-emission pathways from biochar decomposition are accounted for prior to issuance - carbon storage losses calculated using decay-based persistence model. Minimum threshold for $H/C_{org} < 0.7$. End-use application durability: Point of credit creation is at demonstrated durable end-use and prescribes application-specific evidence. For long-lived construction materials, end-of-life pathways are assessed and considered unlikely to threaten carbon. Cascading uses to be documented with associated risks assessed.	Strengths: H/C_{org} threshold ensures minimum biochar carbonisation requirements and thus resistance to decompositions from an assurance of high aromaticity. Issuance only after verified durable use reduces diversion/combustion risk pre-issuance. It demonstrates high specificity and rigour in addressing pathway-specific risks of reversal. Documentation of cascading uses proactively ensures that multi-stage use pathways still result in long-term carbon storage. Aligns incentives by withholding issuance until the physical use is evidenced, effectively pushing handling risk before issuance. Puro's insistence on incorporation materially reduces volatility, combustion exposure, and displacement risk. The methodology also provides an explicit justification, citing peer-reviewed studies showing biochar has low susceptibility to fire once mixed. Risks: Non pathway-specific decay modelling for non-soil applications of biochar. More conservative standards cap thresholds for H/C ratio at 0.4 to eliminate likelihood of semi-persistent carbon fractions in the biochar. While soil disturbance risks are deemed immaterial once biochar is incorporated, certain land-use contexts, such as heavy tillage systems, erosion-prone terrain, or future land conversion, could pose material reversal risks which are not explicitly controlled by the methodology and rely on project and auditor judgement.	
	Misuse risk: Requires robust proof of end-use and extra evidence where misuse risk is higher with well-defined end-use application eligibility requirements. Product handling and storage risk: Combustion during storage/transport classified as a pre-issuance re-emission risk mitigated by evidence of actual end-use. Land-use change and disturbance: For agricultural/forest soils, fires are considered not material once biochar is well incorporated. Mixing/incorporation is required by use rules to reduce exposure.		

Permanence findings – Puro

Puro demonstrates strong safeguards to assure long term durability of carbon sequestration, with some risks arising from limited end-use monitoring requirements.

RAG



Parameters	Puro Biochar 2025 V1 Summary	Strengths and risks	RAG
Monitoring	Requires annual audited monitoring reports covering production volumes, biochar quality parameter, and confirmed end-use applications. Issuance is contingent on documented evidence of final use, ensuring that biochar is already incorporated into soil or durable materials before credits are generated. Monitoring also requires traceability across supply chains to safeguard against diversion. However, once credits are issued, there is no ongoing post-issuance monitoring obligation on the end-use site. Liability for reversal lies with the carbon removal supplier in the event of any reversals.	Strengths: Permanence risks are shifted to pre-crediting stage which reduces exposure to post-credit reversal. Annual audits by accredited third parties and mandatory documentation of end-use represent best practice. Risks: Manual MRV framework provides governance strength, though reliance on documentation and auditor judgement introduces some variability compared to automated dMRV systems. Monitoring does not extend beyond the issuance point, with no requirements for continuous oversight of the storage environment over time. Disturbance events could occur without detection unless voluntarily disclosed by suppliers.	

Permanence findings – Verra



Verra provides some safeguards to ensure durability through eligibility requirements and end-use verification, though post-issuance and pathway-specific risks are not addressed in detail.

RAG

Parameters	Verra VM0044 v1.2 Summary	Strengths and risks	RAG
	Biochar stability and decay risk: Minimum threshold for $H/C_{org} < 0.7$. Projects must adhere to recognized IBI or EBC production and testing standards to confirm chemical recalcitrance and contaminant safety. End-use application durability: Credit issuance is only at the point of verified end-use. For soil applications, mandates that surface-applied biochar be mixed with substrates such as compost or manure to reduce combustibility and erosion. Also excludes application in wetlands. For non-soil applications, only high-technology-produced biochar is eligible, typically embedded in construction materials considered durable carbon sinks. Does not establish procedures to address potential cascading uses. Concludes that overall reversal risks can be considered negligible. Misuse risk: Requires proof of final end-use application through geographic coordinates and chain-of-custody documentation. Product handling and storage risk: Biochar must be applied within one year of production to prevent pre-application decay. Land-use change and disturbance: Natural-disturbance risk is addressed through application-specific management guidance. For soils, biochar must be incorporated into the subsurface (≥ 10 cm) or mixed with other substrates to reduce exposure to fire and erosion.	Strengths: By requiring conformance with IBI/EBC quality protocols, it imports credible third-party standards for pyrolysis control, carbonization completeness, and contaminant screening. The one-year utilization limit effectively caps storage-related oxidation risk before crediting. Mandated mixing or subsurface application helps to minimise risks of decay or combustion. Risks: Relative lack of detail in addressing application-specific risks, relying on limited end-use eligibility criteria. More conservative standards cap thresholds for H/C ratio at 0.4 to eliminate likelihood of semi-persistent carbon fractions in the biochar. Guidance on demonstrating and verifying long-term durability is limited. The evidence requirement for non-soil uses is qualitative, relying on literature or laboratory data without standardized durability testing or defined minimum lifetimes. It does not require ongoing verification of carbon persistence, site-specific assessment of disturbance exposure, or adaptive management plans. There is also no mandate for periodic verification of land-use change or disturbance. Projects can mitigate these risks by adopting end-use application site monitoring such as voluntary long-term sampling or asset stewardship.	
Risks Assessment			

Permanence findings – Verra



Verra provides some safeguards to ensure durability through eligibility requirements and end-use verification, though post-issuance and pathway-specific risks are not addressed in detail.

RAG

Parameters	Verra VM0044 v1.2 Summary	Strengths and risks	RAG
Monitoring	Requires detailed monitoring plans consistent with ISO 14064-2 , including annual data collection on production volumes, carbon content, production temperature, and H/C ratio. Projects must document final end-use applications with geographic coordinates and chain-of-custody records. QA/QC procedures, data management systems, and periodic equipment calibration are also required. No requirement for ongoing post-issuance tracking of biochar persistence or end-of-life conditions.	Strengths: Comprehensive pre-issuance MRV framework captures all relevant production and utilization data, supported by accredited laboratory analysis and QA/QC processes. Risks: Manual MRV framework provides governance strength, though reliance on documentation and auditor judgement introduces some variability compared to automated dMRV systems. Monitoring does not extend beyond the issuance point, with no requirements for continuous oversight of the storage environment over time. Disturbance events could occur without detection unless voluntarily disclosed by suppliers.	

Safeguards and co-benefits

Overview of Sylvera’s approach to assessing safeguards and co-benefits

Pillar	Parameter Assessed	Why it's important
Safeguards and co-benefits	Community safeguarding	Assessing how robust community safeguards are, including the risk of minimum safeguards not being met based on the community engagement & Free, Prior and Informed Consent (FPIC). Robust requirements are desired to ensure no net harm towards local communities.
	Environmental safeguarding	Assessing the risk of minimum safeguards not being met based on consideration of ecosystem services, hydrological, local ecosystems and evidence provided of implementation. Robust requirements are desired to ensure no net harm towards the environment.
	Sustainable development goals (SDGs)	Guidance on how to assess and quantify SDG impact as well as stipulations on minimum contributions can set the floor on co-benefits delivered by a project and confidence in the impact. Desirable that projects demonstrate voluntarily contributions to SDGs.

Safeguarding findings – Isometric

ISOMETRIC

Overall, Isometric sets clear safeguards with strong community protections but offers limited biochar-specific environmental guidance and remains conservative on co-benefits and SDG ambition.

RAG

Parameters	Isometric Biochar Protocol v1.2 Summary	Strengths and Risks	RAG
Community Safeguarding	Requires initial and ongoing stakeholder engagement through a formal Stakeholder Input Process (SIP) covering project planning, implementation, and operations. Where activities may impact or displace Indigenous Peoples or Local Communities, Free, Prior, and Informed Consent (FPIC) is explicitly required. 'Do no net harm' to environment or communities principle enforced.	Strengths: Early and ongoing mandatory stakeholder engagement with grievance mechanisms ensuring a continuous channel for engagement. Mandatory inclusion of stakeholder processes in PDDs aids auditable traceability during validation and verification. Risks: No mechanism for equitable benefit-sharing or social co-benefit delivery is included, meaning communities may be consulted but not necessarily advantaged by the project. The absence of mandatory social impact indicators risks uneven implementation across projects. The methodology does not address biochar-specific social risks.	
Environmental Safeguarding	Mandatory environmental risk screening and harm-prevention requirements across all project stages, including feedstock sourcing, pyrolysis, storage, and end-use. Pollutant screening/limits incl. Polycyclic Aromatic Hydrocarbons (PAHs), heavy metals, polychlorinated biphenyls (PCBs), dioxins/furans with adherence to local/EU/US EPA thresholds. The methodology prohibits the conversion of natural or high-conservation ecosystems and mandates avoidance or minimization of biodiversity harm.	Strengths: Explicit and preventive in relation to contamination. Risks: EIA is encouraged but not mandatory if not legally required in the project jurisdiction. No explicit treatment of biochar-specific risks, such as land-use pressures, soil biodiversity impacts, or long-term ecological disturbance. Reliance on project-level assessment without prescribed methods risks inconsistency.	

Safeguarding findings – Isometric

ISOMETRIC

Overall, Isometric sets clear safeguards with strong community protections but offers limited biochar-specific environmental guidance and remains conservative on co-benefits and SDG ambition.

RAG

Parameters	Isometric Biochar Protocol v1.2 Summary	Strengths and Risks	RAG
Sustainable Development Goals (SDGs)	Projects under Isometric must demonstrate how their carbon removal activities are consistent with relevant SDG objectives. There is no minimum contribution defined in the Isometric Standard nor the Biochar Protocol and there is no guidance on which tools should be used to quantify SDGs.	Risks: Since there is no minimum contribution defined and no guidance on which tools that project developers should be using, this allows for arbitrary impact assessments that can be manipulated. Best practice would be to set a minimum requirement and include tools for SDG measurement.	

Safeguarding findings – Puro

Puro provides strong guidance and requirements on community and environmental safeguarding including compliance with just transition principles

RAG 

Parameters	Puro Biochar 2025 V1 Summary	Strengths and Risks	RAG
Community Safeguarding	Requires initial and ongoing stakeholder engagement with a public Stakeholder Engagement Report and continuous grievance/feedback logging. Free, Prior, and Informed Consent (FPIC) required when activities impact Indigenous Peoples/Local Communities. Puro General Rules align with IC-VCM safeguards: human rights, labor rights (no forced/child labor), non-discrimination and gender equality. Methodology requires statutory EIA/permits where applicable plus a Puro Environmental & Social Safeguards Questionnaire made public after audit. 'Do no net harm' to environment or communities is explicitly required. Mitigations are required to be identified and implemented proportional to risks.	Strengths: Early and ongoing mandatory stakeholder engagement with grievance mechanisms ensuring a continuous channel for engagement and a public report prioritises accountability and transparency. This is reinforced by auditable documentation requirements in the environmental and safeguards questionnaire. Addresses biochar-specific key socio-environmental operational risks with corresponding mitigation measures or requirements applied. Risks: EIA is encouraged but not mandatory if not legally required. Biochar-specific risks addressed prevent harm at operational level, but does not cover social equity topics. Procedural depth for stakeholder consultation is not defined. There is no explicit framework for benefit-sharing.	
Environmental Safeguarding	Standard requires biodiversity conservation, water/soil protection, and minimizing community health/safety impacts. Addresses biochar-specific risks through operational criteria designed to minimise ecological harm across the biochar value chain, e.g. biomass sourcing. Require compliance with national environmental regulations and statutory EIA/permits where applicable. Robust rules for contaminants and hazardous materials across sourcing and production: declare and manage micropollutants/biological hazards, report impurities and restrict plastics, and control air emissions/residues.	Strengths: Detailed, precautionary rules with mandatory reporting and treatment thresholds; explicit handling of biohazards and emissions. Accounts for potential environmental hazards at each project stage such as feedstock sourcing, processing, handling, transport, and end use, are paired with required controls or mitigations. Risks: EIA is encouraged but not mandatory if not legally required.	

Safeguarding findings – Puro

Puro provides strong guidance and requirements on community and environmental safeguarding including compliance with just transition principles

RAG 

Parameters	Puro Biochar 2025 V1 Summary	Strengths and Risks	RAG
Sustainable Development Goals (SDGs)	Voluntary, disclosure-based approach, requiring projects to report intended Puro SDG Attributes (quantifiable, certifiable co-benefits) or SDG Descriptors (qualitative, informational claims). SDG claims are validated ex-ante during the facility audit and verified ex-post during output audits once evidence becomes available. Quantified Attributes must align with IC-VCM CCP Attribute 3 standards and are attached to CORCs only after verification. Projects are not required to meet minimum SDG contribution thresholds or demonstrate tangible social or economic gains.	Strengths: Structured framework promotes transparency and encourages projects to demonstrate positive sustainable development impacts beyond carbon removal. The separation between Attributes and Descriptors provides a consistent structure for communicating co-benefits and enables third-party verification of measurable outcomes. The reference to IC-VCM criteria enhances credibility where projects pursue certified SDG Attributes. Risks: SDG engagement is entirely voluntary and may rely on self-reported, narrative Descriptors without requiring quantified evidence or full verification. Only projects pursuing Attributes must demonstrate measurable results, meaning many projects may report SDG contributions without quantification or third-party evidence. The absence of minimum SDG requirements allows projects to participate without any obligation to deliver or demonstrate broader social or environmental co-benefits.	

Safeguarding findings – Verra



Verra’s methodology outlines a transparent safeguarding and co-benefits mechanism but could be improved by addressing more project-type specific risks.

RAG

Parameters	Verra VM0044 v1.2 Summary	Strengths and Risks	RAG
Community Safeguarding	All Verra methodologies follow the VCS Standard, in which projects are required to implement sustainable development practices, robust stakeholder engagement requirements aligned with FPIC principles and assessments of any potential socio-economic risks that could arise. Risks include labor rights, health and safety, gender impacts, Indigenous livelihoods, and resource access. The Standard has a detailed requirements and guidance section on stakeholder engagement, including steps and timelines for project developers to follow. There are explicit benefit-sharing rules.	Strengths: Verra embeds a rights-based, auditable process. Elements such as FPIC, benefit-sharing agreements, public consultation, ongoing engagement, and grievance mechanisms are transparent and accessible to the general public (when applicable), giving communities multiple entry points to raise and resolve disputes. Risks: No project type-specific guidance is given, which could lead to uncertainty on how to properly address biochar-specific risks.	
Environmental Safeguarding	All Verra methodologies follow the VCS Standard, in which projects are required to include environmental safeguarding rules and requirements, preventing unintended negative consequences to the ecosystem. Considerations of ecosystem services, threatened habitats, risks of invasive species, and hydrological functions of the native ecosystem are included. The use of hazardous or contaminated waste is prohibited, and biochar quality must meet IBI or EBC standards, which impose limits on heavy metals, PAHs, and other contaminants.	Strengths: Verra's "no net harm" environmental safeguards mandate that projects must identify, avoid, and mitigate potential adverse impacts on the environment. It is mandatory for project developers to design and implement measures to mitigate identified risks. Risks: Provides no explicit guidance and threshold on biochar-specific risks, leaving potential gaps where operational or ecological risks differ across project scales or environments.	

Safeguarding findings – Verra



Verra’s methodology outlines a transparent safeguarding and co-benefits mechanism but could be improved by addressing more project-type specific risks.

RAG

Parameters	Verra VM0044 v1.2 Summary	Strengths and Risks	RAG
Sustainable Development Goals (SDGs)	An SDG Impact Assessment is required under the VCS Standard, and any project listed on the Verra Registry must contribute to at least three SDGs. Verra does not outline specific criteria or tools for measuring sustainable development goal contributions.	Strengths: The requirement of the Impact Assessment and the mandated three SDG contributions ensure that projects have positive sustainable development impacts. Risks: Verra does not give clear guidance on how projects should indicate contributions to SDGs, which might cause inconsistent reporting, including over and understating of contributions.	

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Verification Costs and Timeline

Verra is cheapest but is registry-only, Puro is a premium player, Isometric is untransparent

	Isometric	Puro.earth	Verra
Account fees		Annual maintenance fees: €1,400 p.a.	One-time opening fee: US\$750 Annual maintenance fees: US\$750 p.a.
Basic service/issuance fees	Isometric claims to "never take money from suppliers" and only charge fees to buyers. The verification fee is paid by the buyer of the removal credits, not by the project developer. Buyers pay an agreed flat fee for verifying a set amount of removal, regardless of how many credits they ultimately receive.	Annual service fees (estimated): €19,890.00¹ Puro's service fee is a fixed amount per credit depending on the reported cumulative volume of credits issued over the past year. For Astra in the target scenario, this would be a fee of a €10² per credit issued.	Annual issuance fees (estimated): US\$472.42 Verra's issuance fee is a fixed amount of US\$0.23 per tCO₂ of reductions or removals.
Additional service/ Issuance fees	Isometric believes that when buyers pay for verification, it reduces incentive for trade-offs between a registry's bottom line and their scientific integrity.	Each SDG that the supplier would like verified will result in an additional 2.5% premium on top of the total service fee.	Each pipeline listing request costs US\$1,500 Each project registration request costs US\$3,750
Verification/validation fees	However, suppliers may incur indirect costs from choosing to register with Isometric; one example is that buyers might negotiate lower credit purchase prices or contract terms that discount the supplier, because the buyer may factor in the fact that they are bearing the registry costs.	Included in the service fees	Each verification review request costs US\$5,000 , which is composed of a US\$2,500 verification review request fee and a US\$2,500 prepayment fee which can be deducted from future issuance fees owed to Verra).
Additional ad-hoc fees		Optional: €15,000 fee for Puro's Preliminary Assessment service, which is an optional service that provides support to early-stage projects struggling to demonstrate eligibility and who find it difficult to certify.	Each requantification review costs US\$10,000 , applicable if a project requests a review of changes in quantified emissions reductions/removals. Additional fees may apply if a developer requests a new methodology/module or a revision to an existing methodology/module.

(1) Payable when the credit is first traded. Suppliers who pay at the time of issuance can benefit from a 5% discount.

(2) For the full list of service fee tiers, please refer to Puro.earth's [fee schedule](#).

Verification Costs and Timeline

Both Puro.earth and Verra have lengthy timelines for project registration and credit issuance

	Isometric	Puro.earth	Verra
Timeline	Isometric claims that through the registry's monthly verification methods (through the online platform, Isometric Certify), suppliers can deliver credits monthly and realise revenue 10x faster than with other registries. According to Isometric, it takes only a month on average to start issuing credits and just weeks to develop new protocols.	Puro's certification journey involves seven mandatory steps and one optional one: 1. Initial qualification 2. Platform agreement signature 3. Facility registration 4. (Optional) Preliminary assessment 5. Audit preparation 6. VVB facility audit 7. VVB output audit 8. Credit issuances In 2023, Puro's CEO stated that the minimum time frame for a supplier to issue a credit on Puro's registry is 3 months, but can take several years depending on project maturity and the thoroughness of the review process. However, as of October 2025, Puro has rolled out MyPuro 2.0, a refined version of its platform for streamlining certification and reduce the time-to-market for developers. Effectiveness of the platform has yet to be documented.	There are two timelines that project developers need to be aware of: 1. Timeline for registration: After a project proponent submits the draft project description, it undergoes a 30-day public comment period. After this, the project goes through validation in which a VVB reviews the PDD and the comments received during the public comment period. The length of the process varies from project to project; it can take up to a year or longer (in rare cases). Since Verra launched a new approach to this process in November 2024, the timelines have decreased substantially. The current average time for a project to spend with Verra while registration and/or verification requests are being processed was 43 days in the fourth quarter of 2024, compared to 81 days before the new process was implemented. 2. Timeline for issuance: Before a project can issue credits: project activities need to be implemented, the reductions/removals need to be monitored. a VVB needs to verify the ERs that the project achieved, and Verra needs to approve the verification. The exact timeline for this process varies from project to project.

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Market acceptance and industry standards

The lack of a universally accepted criteria for high-quality carbon standards, methodologies, and credits has led multiple stakeholder groups to draft their own principles. Analyzing whether a standard or methodology aligns with the carbon credit supply principles set by these leading groups is a key way to gauge its adherence to industry standards and confirm its market acceptance.

Industry standards that determine market acceptance

ICROA

[Carbon Crediting
Endorsement
Programme](#)

- Launched date: 2008
- The International Carbon Reduction and Offset Alliance (ICROA) is an industry association of carbon market players
- ICROA developed a programme to indicate which carbon standards meet its criteria and the ICROA endorsement list is used voluntarily by many carbon market players since 2008
- ICROA Accredited Organisations (incl. major intermediaries and project developers) have committed to only use ICROA-endorsed standards

CORSIA

[Eligible Emission Units
\(EEUs\)](#)

- Launched date: 2019
- The Carbon Offset and Reduction Scheme for International Aviation (CORSIA) is the international compliance scheme regulating emissions for international civil aviation and it is managed by International Civil Aviation Organization (ICAO), a UN agency.
- CORSIA has approved carbon standards and methodologies that can supply CORSIA Eligible Emission Units (or carbon credits) to airlines, but it is often used as a benchmark outside the aviation sector as well.

ICVCM

[Core Carbon
Principles \(CCPs\)](#)

- Launched date: 2021
- The Integrity Council for Voluntary Carbon Markets (ICVCM) is an independent governance body for the voluntary carbon market that aims to set global benchmarks for carbon credits through its Core Carbon Principles (CCPs).
- The CCPs are the emerging leading industry quality standard and are likely to be seen as a quality threshold within and beyond the VCM.

Compliance carbon pricing schemes*

- Compliance carbon pricing schemes have existed in different regions and countries since the early 2000s.
- [40% of the schemes allow](#) the use of carbon credits for corporate compliance towards emissions limits, but these are often subject to restrictions based on activity type, vintages, geography and more.

* Credits no older <3 years than the emissions being offset

Market acceptance and industry standards

Overall, the comparison indicates that Verra currently benefits from the widest formal recognition, while Isometric and Puro are progressing through conditional endorsements and ongoing ICVCM assessments, reflecting an evolving market landscape for Biochar methodologies.

Industry standards		Puro Biochar Methodology for CO2 Removal Edition 2025 V1	Verra VM0044 Methodology for Biochar Utilization in Soil and Non-Soil Applications, v1.2	Isometric Biochar Production and Storage V1.2
ICROA Carbon Crediting Endorsement Programme	Standard	Puro is ICROA endorsed	VCS is ICROA endorsed	Isometric is ICROA conditionally endorsed
	Methodology	n/a	n/a	n/a
CORSIA First Phase Eligible Emission Units (EEUs)	Standard	Puro didn't apply to be CORSIA approved	VCS is CORSIA approved	Isometric is CORSIA conditionally approved
	Methodology	n/a	The methodology is currently ineligible because it did not exist when CORSIA approved the VCS as a standard for the pilot or initial phases. Furthermore, Verra has not yet submitted a Material Change form to request the inclusion of this methodology.	n/a
ICVCM Core Carbon Principles (CCPs)	Standard	Puro is not yet CCP-eligible, it is scheduled for "Governing Board consideration"	VCS is CCP-eligible	Isometric is CCP-eligible
	Methodology	V3 assessment in progress	V1.2 is CCP-approved	• V1.0 is CCP-approved and • V1.1 is going through assessment. • V1.2 is not finalized and has not applied for CPP label yet
Compliance carbon pricing schemes*	Standard and methodology	Not eligible	Projects located in Chile*, South Africa** and Rwanda*** are likely to be eligible under the Chile and the Singapore carbon tax respectively but conditions apply. Projects located in Colombia could be eligible under the Colombia carbon tax	Not eligible

* Credits no older <3 years than the emissions being offset

** Pending extended letter of approval

***Pending Joint-Committee approval and corresponding adjustments

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Biomass Sourcing Criteria – Sustainability requirement and certifications

Feedstock Type	Example Feedstock types & Point of Origin	Puro	Isometric	Verra
Waste, forest product	- Off-cuts, sawdust → Sawmill/processing facility - Thinnings from wildfire reduction → Forest harvest site - Wood processing waste, bark → Processing facility - Forest management residues → Forest harvest site - Wood pallets, construction waste → Secondary processing/demolition sites	Puro.earth Biomass Sourcing Criteria 2025 V11. (Category G, H): - Eligible if residues or by-products from forest processing (e.g., sawdust, pulp sludge). - Sustainability evidence options: recognized forest/biomass sustainability certifications (FSC, RSB, SBP, SFI, PEFC, others up to Puro.earth determination); compliance with EU RED III or equivalent national rules under UNFCCC; government-approved local forest management plans (Corruption Perception Index (CPI) ≥ 50); OR third-party audited supplier data verified in line with ISAE 3000 limited assurance engagement or equivalent as a minimum. - Certification not strictly required if traceability (origin, supplier, records) is sufficient, but some structured evidence must exist.	Isometric Biomass Feedstock Accounting, v1.2 (Session 2.1 and Appendix 3): - Must satisfy market leakage criteria - Acceptable certifications: FSC, PEFC, SFI, ATFS - Requires documentation that forest carbon stocks are stable or increasing over last 5–10 years - Chain of custody documentation: purchase records, forest management plans, NEPA or equivalent - Additional evidence required if using SBP, RSB, FSC Mix (limited to 50%)	Verra VM0044 (Table 1, page 10): - Must prove biomass comes from sustainable sources and does not lead to deforestation - Example proof of sustainable sourcing (FSC, PEFC, or government-approved plans)
	- Crop residues (straw, stalks, leaves) → Farm/field where harvested - Husks, shells, pomace → Primary processing mills - Food processing waste → Food processing facilities - Expired food products → Food manufacturing/retail facilities - Coffee, tea, cocoa residues → Beverage processing plants	Puro.earth Biomass Sourcing Criteria 2025 V11 (Categories B, D, K, L, E, F): - Certifications not required—basic traceability and regulatory compliance are sufficient. - Evidence options: primary evidence from the supplier regarding legal operations and working conditions where required; delivery records, supplier and geographic sourcing info; permits for hazardous fractions.	Isometric Biomass Feedstock Accounting, v1.2 (Session 2.1 and Appendix 3): - For documentation, must meet EC6 (indirect) and EC11–EC12 (direct) where applicable - Must document counterfactual fate and potential CH₄/CO₂e emissions for each batch (EC13) - If material serves no other function (e.g., stockpiles), replacement emissions = 0 (Section 3.2.2) - Fertilizer replacement emissions must account for equivalent NPK via lab or literature-based nutrient content	Verra VM0044 (Table 1, page 10): - Project proponents must provide documentation that the project activity is not leading to a decline in soil carbon stocks or a reduction in crop productivity, or that in the baseline agriculture waste biomass was burned without energy production (e.g., open burning of stubble). - In the absence of documentation, feedstock removal is limited to no more than 50 percent of total residues to protect against soil degradation

Appendix

Biomass Sourcing Criteria – Sustainability requirement and certifications

Feedstock Type	Example Feedstock types & Point of Origin	Puro	Isometric	Verra
Non-waste, forest product (Generally PROHIBITED or highly restricted)	- Purpose-grown timber → Managed forest plantations - Short rotation coppice → Dedicated energy forests - Whole trees → Forest stands managed for biomass	Puro.earth Biomass Sourcing Criteria 2025 V1.1 (Category G): - Purpose-grown forest products (e.g., plantation timber) highly restricted. - Evidence options: forest sustainability certifications (as above), strong regulatory compliance (EU RED III, national rules), or third-party verified management plans. - Without structured certification/regulation, typically ineligible due to criteria on regeneration, carbon stocks, biodiversity, and exclusion of high-quality timber.	Isometric Biomass Feedstock Accounting, v1.2 (Session 2.1): - Prohibited unless biomass is from mandated wildfire mitigation or similar programs (EC14) - Must be certified under FSC, PEFC, SFI, or ATFS, with verification of harvest justification - Requires documentation proving biomass would otherwise decompose and release carbon within 15 years	Verra VM0044 (Session 4, page 9): - Feedstock must be purely biogenic waste biomass and not purpose-grown - Feedstock must have been otherwise left to decay or combusted for purposes other than energy production.
Non-waste, non-forest (Generally DISCOURAGED)	- Miscanthus, switchgrass → Dedicated energy crop fields - Bamboo plantations → Managed bamboo forests - Hemp, kenaf → Agricultural fields - Algae → Cultivation ponds/bioreactors - Invasive species → Areas requiring ecological restoration	Puro.earth Biomass Sourcing Criteria 2025 V1.1 (Category I): - Dedicated crops eligible only on degraded, marginal, or contaminated land officially recognized by competent authorities; cover or intermediate crops on agricultural land (i.e., not displacing food/feed). - Evidence options: agricultural sustainability certifications (RSB, ISCC, RED III compliance); Land-use maps, registers, or other official documentation showing degraded/marginal land or cover crop status; Supplier-provided primary evidence on sustainable cultivation practices (soil protection, water use, biodiversity). - Certification not mandatory, but robust land-use documentation and sustainable practice evidence required.	Isometric Biomass Feedstock Accounting, v1.2 (Session 2.1) Dedicated feedstocks only eligible if: - Not grown for energy use (EC15) - <50% of local biomass (by value/volume) is used for energy, or supplier has no history of selling to ethanol plants - Project must provide spatial analysis or local market documentation	Verra VM0044 (Session 4, page 9): Purpose-grown & energy crops avoided

Appendix

Registries & Documentation Sources

Isometric Registry

- [Biochar Production and Storage V1.2](#) - Under public consultation until 10/11/2025
- [Biomass Feedstock Accounting V1.3](#) - Certified on 16/10/2025
- [Isometric Standard V1.7.6](#) - Certified on 28/08/2025
- [GHG Accounting V1.0](#) - Certified on 21/08/2025
- [Embodied Emissions Accounting V1.0](#) - Certified on 24/04/2024
-

Puro Standard Registry

- [Biochar Methodology for CO2 Removal 2025 V1.0](#) - Published on 12/06/2025
- [Puro Standard General Rules V4.2](#) - Published on 31/03/2025
- [Additionality Assessment Requirements V2](#) - Published on 07/06/2024
- [Biomass Sourcing Criteria V1.0](#) - 30/05/2024

Verra VCS Registry

- [VM0044 Biochar Utilization in Soil and Non-Soil Applications v1.2](#) - Published on 27/06/2025
- [VCS Standard Edition 2024 V4.7](#) - Published on 16/04/2025
- [VT0008 Module for Additionality Assessment V1.0](#) - Published on 14/10/2024

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